

Honors College Thesis Proposal

Does aging microplastics alter how microplastics affect the physiology of a freshwater copepod

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Thesis Details

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Project Format: Research Thesis

Project Output: Article length written paper

Proposal Background

Microplastics are a growing concern in the world today. The amount of these microplastics has been increasing over the years (REF) , but not much is known about their long term effects on living organisms. Microplastics have been observed to acquire biofilms of microscopic organisms such as bacteria and algae while in the environment (Ballesté et al., 2024).

Copepods are small zooplankton are some of the most important organisms in the food chain as they consume phytoplankton and they themselves are a very abundant, protein rich food source for animals as small as fish larvae to whales. Previous studies with marine copepods have shown a potential hormetic curve with their CTmax when consuming a small quantity of microplastics (Zhang et al., 2025) . Other studies have also pointed out that copepods are able to detect microplastics and avoid them when feeding with a success rate of up to 80% (REF) , even in the presence of a biofilm (Xu et al., 2022). These dynamics and the effects of biofilms may, however, vary across species (Fagiano et al., 2025).

The effect of particle aging and biofilm formation may also depend on the nutritional state of the copepod. Under low-food conditions, the presence of a biofilm may provide some nutritional benefit to grazers (Botterell et al., 2020), thus reducing the selective feeding behaviour and increasing exposure to microplastics.

Research Questions and Hypotheses

This research project will examine the effects of microplastic exposure on copepod physiology, and test whether these effects are altered by aging in environmental water samples (i.e. promoting the formation of a biofilm). The question being addressed is: Does aging microplastics alter how microplastics affect the physiology of a freshwater copepod?

Question 1: Does exposure to microplastics affect upper thermal limits in the freshwater copepod *Skistodiaptomus pallidus*?

Hypothesis 1: XYZ

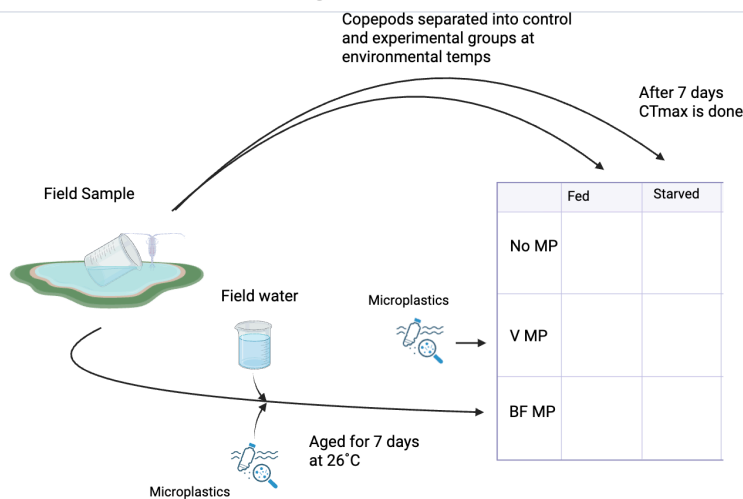
Question 2: Does the effect of microplastics on upper thermal limits differ between aged and un-aged microplastics?

Hypothesis 2: XYZ

Question 3: Does starvation alter the effects of microplastics on upper thermal limits?

Hypothesis 3: XYZ

Experimental Design



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Hypothesis Testing

We will be using a mixed effects model to test our hypotheses. We are testing three hypotheses.

For our hypothesis that XYZ we expect to see ABC if the hypothesis is supported .

Project Timeline

Month	Progress
September	Background reading, Planning, 1st experimental replicate
October	Experimental replicates, prepare poster on preliminary data
November	Final Experimental Replicates, Data analysis
December	Data analysis
January	Writing thesis
February	Writing thesis
March	Submit draft thesis to committee member, Prepare poster for student research and community engagement symposium
April	Writing thesis, Prepare final presentation for honors program
May	